Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider two discrete memoryless sources X and Y, whose joint p.d.f. is given by

	b_1	b_2	$p(a)$
a_1	0.8	0.02	0.82
a_2	0.03	0.15	0.18
$p(b)$	0.83	0.17	1.0

   (a) Determine whether X and Y are statistically dependent sources.

   (b) Compute a lower bound on the average length of a code used to encode: i) X and Y independently; ii) X and Y jointly.

   (c) Consider the following sequences of symbols

source	symbols
X	a_1, a_1, a_2, a_1
Y	b_1, b_1, b_2, b_2

   An arithmetic encoder is designed to encode the symbols emitted by the source X, followed by those of the source Y. Compute the number of bits used by such arithmetic coder in two cases: i) neglecting, or ii) exploiting the statistical dependency between the sources.

2. **Question (5 pts)**
   Discuss the design of the filter bank in audio coding.

3. **Question (5 pts)**
   (a) Illustrate the properties of speech signals, both in the temporal and in the frequency domain.

   (b) Describe the source-system model used to synthesize speech signals.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a transform coding system that processes blocks of N input samples.

   - [T] [F] When the KLT transform is used, the transform coefficients are statistically independent.
   - [T] [F] Let $G_{TC}(N)$ denote the transform coding gain achievable using blocks of N samples. Then, $\lim_{N\to\infty} G_{TC}(N) = +\infty$.
   - [T] [F] $R_{TC}(D) \leq R_{PCM}(D)$, where $R_{TC}(D)$ and $R_{PCM}(D)$ are, respectively, the operational rate-distortion curves of transform coding and scalar quantization.
   - [T] [F] The asymptotic time complexity of computing the output of a transform coder is $O(N)$.

   (b) Consider a quantizer based on compander-expander

   - [T] [F] $D_{CE}(R) \leq D_U(R)$, where CE and U refer to, respectively, the compander-expander and uniform quantizer.
   - [T] [F] In a compander-expander quantizer, the reconstruction levels are equally likely.
   - [T] [F] A compander-expander quantizer can be conveniently used to encode speech signals.

   (c) Consider an image coding system.

   - [T] [F] Setting the same quantization step size for all transform coefficients leads to a fixed number of bits for all blocks.
   - [T] [F] Let R denote the rate to encode an image. If $R_2 > R_1$, then $PSNR(R_1) > PSNR(R_2)$.
   - [T] [F] Let R_I denote the rate to encode image I. If $R_I = R_J$, then $PSNR_I(R_I) = PSNR_J(R_J)$.

5. **Question (Matlab - 10pts)**

   Consider the image *image.png*.

   (a) Read the image pixels column-wise into a single vector.

   (b) PCM encode the pixel values.

   (c) DPCM encode the pixel values.

   (d) Plot the rate-distortion function $PSNR(R)$ in both cases by varying the quantization step size. Use the entropy of the quantization symbols as an estimate of the rate.